

Fast falsification. Shorter research loops.

Under human direction, Codex turns proof gaps into C++ tests, attacks weak lemma routes, reads the failures and carries the surviving structure into the next candidate.

PRIMARY ADVANTAGE

Bad directions fail quickly.

- Exact numerical objectives stress a proposed lemma.
- Counterexamples and failure certificates become reusable data.
- AI uses each failure to refine the next proof obligation.

CURRENT ACTIVE LAB

3D Navier–Stokes existence and smoothness

Research code and finite evidence — not a Clay solution.

Seven problems, one active laboratory

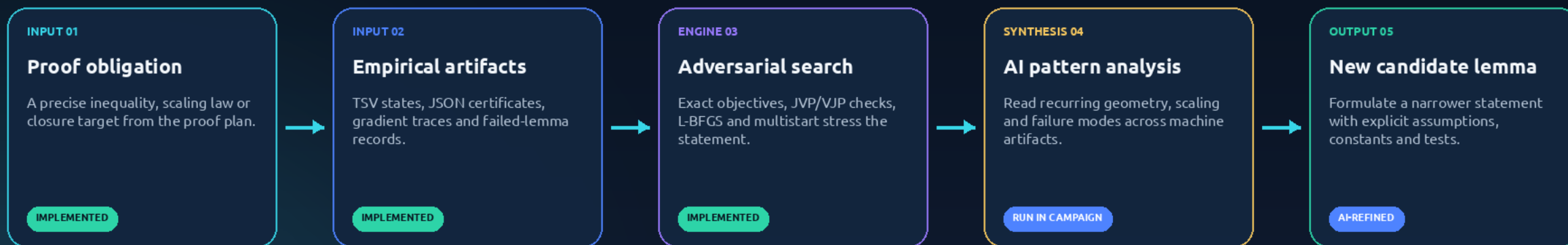
The repository is organized by Millennium Prize Problem. Navier–Stokes is active; the other unsolved tracks are explicit future scope.

PROBLEM	AREA	CLAY STATUS	REPOSITORY
Birch–Swinnerton-Dyer	Number theory	NOT STARTED	Planned
Hodge conjecture	Algebraic geometry	NOT STARTED	Planned
Navier–Stokes	PDE / fluid dynamics	ACTIVE	C++ lemma lab
P vs NP	Complexity theory	NOT STARTED	Planned
Poincaré conjecture	Topology	SOLVED EXTERNALLY	No lab needed
Riemann hypothesis	Number theory	NOT STARTED	Planned
Yang–Mills mass gap	Mathematical physics	NOT STARTED	Planned

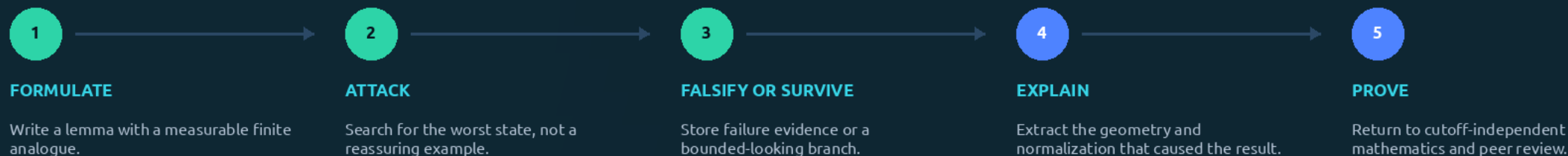
Official problem definitions and current solved/unsolved classification: claymath.org/millennium-problems

AI executes the research loop

Under human direction, Codex formulates candidates, writes and runs the C++, interprets failures and designs the next lemma. Every claim returns to exact tests and expert proof review.



THE HUMAN-DIRECTED, AI-EXECUTED RESEARCH LOOP



Guardrail: a finite certificate may reject a proposed route; it cannot by itself establish a uniform theorem.

The engine already changes which lemmas are worth pursuing

One route was rejected by exact adversarial evidence; the surviving route became a narrower joint normalization problem.

FALSIFIED ROUTE

PNT-13: standalone shell decorrelation

The proposed exponential height-gap decay is contradicted by optimized finite states with near-perfect normalized shell correlation.

NORMALIZED CORRELATION



RESEARCH TIME SAVED

Stop trying to prove correlation decay in isolation. Preserve the full coupled normalization instead.

SURVIVING TARGET

PNT-12: joint normalized tail bound

The direct exact-gradient objective keeps the Gram row, tail mass and common normalization coupled. This route is still open.

EVIDENCE	RESULT	MEANING
H16, K12	4.18215e-4	New finite stress record
Gradient check	1.21e-10	Directional error
Peak RSS	4.88 GiB	Down from 7.94 GiB

STATUS Bounded-looking finite evidence; no uniform proof yet.

A useful engine is honest about the distance to a proof

The project has produced exact reductions, a proved partial tail lemma and strong falsification tooling. The central closure chain remains open.



NEXT ENGINE MILESTONES

WORK ITEM	PURPOSE	EXIT CONDITION
Height/cutoff scans	Stress PNT-12 row by row	Stable or growing branch classified
AI artifact schema	Compare JSON/TSV evidence	Ranked candidate lemmas with provenance
Analytic restart	Prove joint normalization	Cutoff-independent inequality

All machine claims remain reproducible from repository commands and certificates.

NON-CLAIM

The Clay problem is not solved.

A proof still requires cutoff-independent analysis, the full local closure, the complementary normalization channel, and the remaining L4–L6 argument.

RESEARCH SOFTWARE

Evidence generator • falsifier • proof navigator